VSIT | Vidyalankar School of Information Technology
RE-ACCREDITED : GRADE 'A' BY NAAC

VITD2301 : Statistical Methods -1
Practical #1

Name	Harshal Ravindra Gardre	**Roll Number**	26302A3005
Class	SYIT	**Division**	A
Subject/Course	VITD2301 : Statistical Methods -1		
Topic	To verify and apply the concepts of conditional probability, event independence, and the addi Microsoft Excel.		

Question 1. What is the probability that the sum of the two rolls is exactly 7 or 11?

	A	B	C	D	E	F
1	S={(1,1),(1,2),(1,3),(1,4),(1,5),(1,6),(2,1),(2,2),(2,3),(2,4),(2,5),(2,6					
2	n(S)	36				
3	A={(1,6),(2,5),(3,4),(4,3),(5,2),(6,1),(5,6),(6,5)}					
4	n(A)	8				
5						
6	P(A)	0.222222				
7						

Question 2. What is the probability that both dice show a number greater than 3?

7		
8	B={(4,4),(4,5),(4,6),(5,4),(5,5),(5,6),(6,4),(6,5),(6,6)}	
9	n(B)	9
10		
11	P(B)	0.25

Question 3. If the first die shows a 6, what is the chance that the second die will be a 4?

12				
13	C={(1,4),(2,4),(3,4),(4,4),(5,4),(6,4)}			
14	n©	6		
15				
16	P©	0.166667		

Question 4. Given that the first die is 5, what is the probability that the sum of the two dice is 8?

17					
18	A={(2,6),6,2),(3,5),(5,3),(4,4)}				
19	n(A)	5			
20					
21	B={(5,1),(5,2),(5,3),(5,4),(5,5),(5,6)}				
22	n(B)	6			
23					
24	A∩B={(5,3)}		n(A∩B)	1	

Question 5. What is the probability that an employee was promoted given they were trained?

24	A∩B={(5,3)}		n(A∩B)	
25	P(B)	0.166667		
26	P(A∩B)	0.027778		
27	P(A|B)	0.166667		
28				

Question 6. What is the probability that an employee was trained given they were promoted?

VSIT | Vidyalankar School of Information Technology
RE-ACCREDITED : GRADE 'A' BY NAAC

VITD2301 : Statistical Methods -1
Practical #1 – Practice Exercise

Name	Shubham Mohite	**Roll Number**	25302A0062
Class	SYIT A	**Division**	A
Subject/Course	VITD2301 : Statistical Methods -1		
Topic	To verify and apply the concepts of conditional probability, event independence, and the addition theorem using Microsoft Excel.		

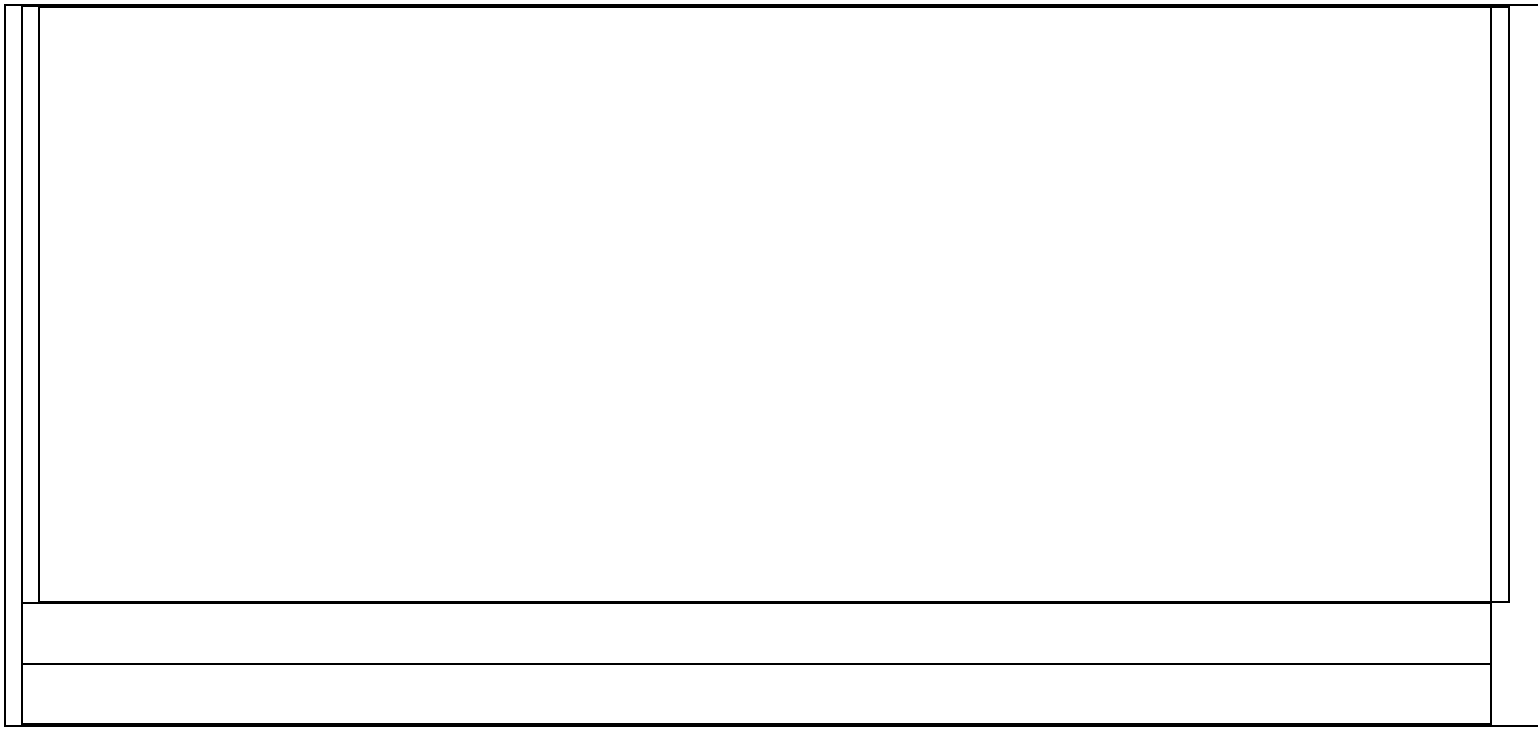
Question 1. Find the probability of getting an even number on a die.

Question 2. Find the probability of getting a prime number on a die.

Question 3. Find the probability of drawing a king from a deck of cards.

Question 4. Explain conditional probability.

Question 5. What are independent events?



VSIT | Vidyalankar School of Information Technology
RE-ACCREDITED : GRADE 'A' BY NAAC

VITD2301 : Statistical Methods -1
Practical #2

Name	Harshal Ravindra Gadre	**Roll Number**	26302A3005
Class	SY-BSC IT	**Division**	A
Subject/Course	VITD2301 : Statistical Methods -1		
Topic	To understand and apply the concepts of Bayes' Theorem, Addition Theorem, Conditional Probability, and Dependent Events using Microsoft Excel.		

Question 1. What is the Probability for a defective product being from either Type A or Type B?

Data:

Product Type	Defective Products
A	12
B	8
Total Defective	20

Question 1. What is the Probability for a defective product being from either Type A or Type B?

Data:		
peoduct		Defective Product
A		12
B		8
Total Defective		20
A	Defective product type A	
B	Defective product type B	
n(A)	12	
n(B)	8	
n(S)	20	
n(A^B)	Number of dcefective product from type A And type B	
	0	
n(AUB)	Number of defective product from type A or type B	
	n(A)+n(B)-n(A^B)	
	12+8-0	
	20	
P(AUB)	n(AUB)/n(S)	
	20/20	
	1	

Question 2. What is the Probability of a Defective Product Being from Type A?

Data:		
peoduct		Defective Product
A		12
B		8
Total Defective		20
A	Defective product type A	
B	Defective product type B	
n(A)	12	
n(B)	8	
n(S)	20	

Q1What is the Probability of a Defective Product Being from Type A?	
let event A be the total defective product	
n(A)=	12
p(A)	n(A)/n(S)
	0.6

Question 3. What is the Probability of a Defective Product Being from Type B?

Data:		
peoduct		Defective Product
A		12
B		8
Total Defective		20
A	Defective product type A	
B	Defective product type B	
n(A)	12	
n(B)	8	
n(S)	20	

Q3. What is the Probability of a Defective Product Being from Type B?	
let event B be the total defective product	
n(B)	8
p(B)	n(B)/n(S)
	0.4

Question 4. What is the Probability that you pass the screening test and get the final interview?

Given:

- P(Screening Pass) = 0.75
- P(Final Interview | Screening Pass) = 0.85

Question 4. What is the Probability that you pass the screening test and get the final interview?

P(Screening Pass) = 0.75	
P(final Pass|Screening pass) = 0.85	
A	Pass the screening test
B	got the final interview
P(A)	0.75
P(B|A)	0.85
P(P^B)	P(B|A)*P(A)
	0.6375

Question 5. What is the Overall probability that a customer has a low credit score?
Given:

- P(Default) = 0.10
- P(Low Credit | Default) = 0.70
- P(No Default) = 0.90
- P(Low Credit | No Default) = 0.20

Question 5. What is the Overall probability that a customer has a low credit score?	
P(Default) = 0.10	
P(Low Credit | Default) = 0.70	
P(No Default) = 0.90	
P(Low Credit | No Default) = 0.20	
A	customer is defaulter
A'	customer is not defaulter
B	customer has low credit score
P(A)	0.1
P(A')	0.9
p(B|A)	0.7
P(B|A')	0.2
P(B|A)*P(A)+P(B|A')*P(A')	
p(Low credit)=P(Low Credit | Default)*P(Default)+P(Low Credit | No Default)*P(No Default)	
	0.25

Question 6. If a customer has a low credit score, what is the probability they will default?

Question 6. If a customer has a low credit score, what is the probability they will default

P(Default) = 0.10
P(Low Credit | Default) = 0.70
P(No Default) = 0.90
P(Low Credit | No Default) = 0.20

A	customer is defaulter
A'	customer is not defaulter
B	customer has low credit score
P(A)	0.1
P(A')	0.9
p(B|A)	0.7
P(B|A')	0.2

P(A|B)=P(B|A)*P(A)

0.07

VSIT Vidyalankar School of Information Technology
RE-ACCREDITED : GRADE 'A' BY NAAC

VITD2301 : Statistical Methods -1
Practical 2 – Practice Exercise

Name	Siddhesh rewale	**Roll Number**	25302A0003
Class	SYBSC- IT	**Division**	A
Subject/Course	VITD2301 : Statistical Methods -1		
Topic	To understand and apply the concepts of Bayes' Theorem, Addition Theorem, Conditional Probability, and Dependent Events using Microsoft Excel.		

1. What is the Probability that an accident happens at night o Question r during rain? Given:

- P(Night)=0.30
- P(Rain)=0.40
- P(Night ∩ Rain)=0.15

1. What is the Probability that an accident happens at night o Question r during rain?

given

let (A) be the probability of accident happen at a night

P(A) 0.3

let B be the probability of accident happen during rain

P(B) 0.4

the probability of accident happen at night and during rain

P(A^B) 0.15

to find P(AUB)=P(A)+p(B)-P(A^B)

0.3+0.4-0.15

0.55

Question 2. If an accident occurred during rain, what is the probability it happened at night?

given

let (A) be the probability of accident happen at a night

P(A) 0.3

let B be the probability of accident happen during rain

P(B) 0.4

the probability of accident happen at night and during rain

P(A^B) 0.15

to find P(A|B)=P(A^B)/P(B)

0.15/0.4

Question 3. What is the Probability that a randomly selected student either attended at least 50 days or participated in extracurricular activities?

Assume:

- Students with Attendance ≥ 50 = A
- Students in Activities = B
- Students satisfying both = A∩B

Question 3. What is the Probability that a randomly selected student either attended at least 50 days or participated in extracurricular activities?

Given

A = Students with Attendence greater than equal to 50

B = Students in Activities

C = Students satisfyimg both

to find P(AUB)=P(A)+P(B)-P(A^B)

Question 4. What is the Probability that a randomly selected student attended at least 50 days and passed the final exam?

A = Students with Attendence greater than equal to 50

B = Students in Activities

C = Students satisfyimg both

to find P(A^B)=P(A)*P(B|A)

Question 5. What is the Probability that a randomly selected student attended at least 50 days and passed the final exam?

Question 5. What is the Probability that a randomly selected student attended at least 50 days and passed the final exam?

let the total no of students be N

A = Students with Attendence greater than equal to 50

B = Students in Activities

C = Students satisfyimg both

to find P(A^B)=P(A)*P(B|A)

VSIT | Vidyalankar School of Information Technology
RE-ACCREDITED : GRADE 'A' BY NAAC

VITD2301 : Statistical Method -I
Practical #4

Name	Harshal Ravindra Gadre	**Roll Number**	26302A3005
Class	SYIT	**Division**	A
Subject/Course	VITD2301 : Statistical Method- I		
Topic	**To understand and apply the concepts of Bivariate Distributions for analysis using Excel**		

Question 1. The joint Distribution:

	Y = 1	Y = 2	Y = 3
X = 0	0.1	0.2	0.1
X = 1	0.2	0.1	0.3

a. **Find total Probability**
b. **Find P(X = 0), P(X = 1)**
c. **Find P(Y = 1), P(Y = 2), P(Y = 3)**

X/Y	Y	Y	Y
X/Y	1	2	3
0	0.1	0.2	0.1
1	0.2	0.1	0.3
Total Probability	1		
P(X=0)	0.4		
P(X=1)	0.6		
P(Y=1)	0.3		
P(Y=2)	0.3		
P(Y=3)	0.4		

Question 2. The joint Distribution:

	Y = 1	Y = 2	Y = 3
X = 0	0.1	0.2	0.1
X = 1	0.2	0.1	0.3

a. **Find expectation of Y, E[Y]**
b. **Find expectation of X, E[X]**

X/Y	Y	Y	Y
	1	2	3
0	0.1	0.2	0.1
1	0.2	0.1	0.3
Total Probability	1		
P(X=0)	0.4		
P(X=1)	0.6		
P(Y=1)	0.3		
P(Y=2)	0.3		
P(Y=3)	0.4		
E[X] = SUM * Probability of X			
E[X]	0.6		
E[Y] = SUM * Probability of Y			
E[Y]	2.1		

Question 3. The joint Distribution:

	Y = 1	Y = 2	Y = 3
X = 0	0.1	0.2	0.1
X = 1	0.2	0.1	0.3

a. **Find joint expectation, E[XY]**

X/Y	Y	Y	Y
	1	2	3
0	0.1	0.2	0.1
1	0.2	0.1	0.3
Total Probability	1		
P(X=0)	0.4		
P(X=1)	0.6		
P(Y=1)	0.3		
P(Y=2)	0.3		
P(Y=3)	0.4		
E[XY]	1.3		

Question 4. The joint Distribution:

	Y = 1	Y = 2	Y = 3
X = 0	0.1	0.2	0.1
X = 1	0.2	0.1	0.3

a. **Find the variance of X, Var(X)**
b. **Find the variance of Y, Var(Y)**

X/Y	Y	Y	Y
	1	2	3
0	0.1	0.2	0.1
1	0.2	0.1	0.3
Total Probability	1		
P(X=0)	0.4		
P(X=1)	0.6		
P(Y=1)	0.3		
P(Y=2)	0.3		
P(Y=3)	0.4		
E[X] = SUM * Probability of X			
E[X]	0.6		
E[Y] = SUM * Probability of Y			
E[Y]	2.1		
E[X^2]	0.6		
X[X]^2	0.36		
Var[X]	0.24		
E[Y^2]	5.1		
E[Y]^2	4.41		
Var[X]	0.69		

Question 5. The joint Distribution:

	Y = 1	Y = 2	Y = 3
X = 0	0.1	0.2	0.1
X = 1	0.2	0.1	0.3

a. **Find the covariance, Cov(X,Y)**
b. **Find the correlation coifficient**

VSIT Vidyalankar School of Information Technology
RE-ACCREDITED : GRADE 'A' BY NAAC

VITD2301 : Statistical Method -I
Practical 4 – Practice Exercise

Name		**Roll Number**	
Class	SYIT	**Division**	
Subject/Course	VITD2301 : Statistical Method- I		
Topic	**To understand and apply the concepts of Bivariate Distributions for analysis using Excel**		

Question 1. The joint Distribution:

	Y = 0	Y = 1	Y = 2
X = 0	0.05	0.1	0.15
X = 1	0.1	0.2	0.1
X = 2	0.05	0.15	0.1

a. Find Total Probability

Question 2. Find P(X) and P(Y = 1)

Question 3. Find Expectation of X and Y. E[X] and E[Y]

Question 4. Find joint Expectation E[XY]

Question 5. Find Variance of X and Y. Var(X) and Var(Y)

Question 6. Find conditional probability. P(X = 1 | Y = 1)

Question 7. Find probability, P(X ≥ 1, Y ≥ 1)

VITD2301 : Statistical Method -I
Practical #6

Name	Harshal Ravindra Gadre	**Roll Number**	26302A0005
Class	SYIT	**Division**	A
Subject/Course	VITD2301 : Statistical Method- I		
Topic	**To compute, visualize, and interpret Poisson probabilities and related statistical measures in understanding the modeling of random events occurring over a fixed interval of time or spac**		

FORMULA = POISSION.DIST(X,M,FALSE)

Question 1. Find probability mass function (pmf) of *P(X = 3)* for parameter 2. Find cumulative probability for *P* probability of *P(X ≥ 2)*.

FORMULA = POISSION.DIST(X,M,FALSE)

P(X=x)	#NAME?
x	3
m	2
P(X=3)	0.180447
P(X<=3)	0.857123
P(X>=2)	1-P(X<=1)
	1-poisson.dist(3,2,true)
	0.593994

Question 2. What is the probability of zero occurrence, with parameter 4.

WHAT IS THE PROBAIBLTY OF ZERO OCCURRENCE , WITH PARAMETER 4	
X	0
M	4
P(X=0)	0.018316

Question 3. Find the mean and variance of the discrete random variable following Poisson distribution with parameter 5. Find the standard deviation.

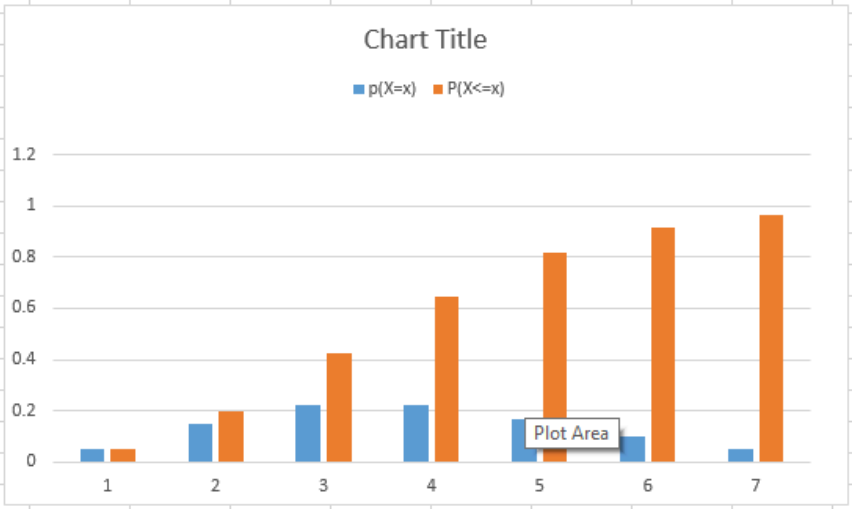
Question 3. Find the mean and variance of the discrete random variable following Poisson distribution with parameter 5. Find the standard deviation.		
m	4	
E(X)	m	4
V(X)	m	4
SD(X)	sqrt(V(X))	2

Question 4. Create probability table: for parameter 3 and x = 0 to 6. Plot a graph for Poisson distribution.

Question 4. Create probability table: for parameter 3 and x = 0 to 6. Plot a graph for Poisson distribution.

M	3
X	1,2,3,4,5,6

X	M	P(X=x)	P(X<=x)
0	3	0.04978707	0.04978707
1	3	0.14936121	0.19914827
2	3	0.22404181	0.42319008
3	3	0.22404181	0.64723189
4	3	0.16803136	0.81526324
5	3	0.10081881	0.91608206
6	3	0.05040941	0.96649146



Question 5. Find $P(2 \leq X \leq 5)$ with parameter 3.

5. Find $P(2 \leq X \leq 5)$ with parameter 3.

m	3
P(X<=1)	0.19914827
P(X<=5)	0.91608206
P(2<=5<=5)	P(X<=5)-P(X<=1)
	0.716934

VITD2301 : Statistical Method -I
Practical 6 – Practice Exercise

Name		**Roll Number**	
Class	SYIT	**Division**	

Subject/Course	VITD2301 : Statistical Method- I
Topic	**To compute, visualize, and interpret Poisson probabilities and related statistical measures in Excel, thereby understanding the modeling of random events occurring over a fixed interval of time or space.**

Question 1. Find $P(X = 4)$ with parameter 3.

Question 2. Find $P(X \leq 2)$ with parameter 5.

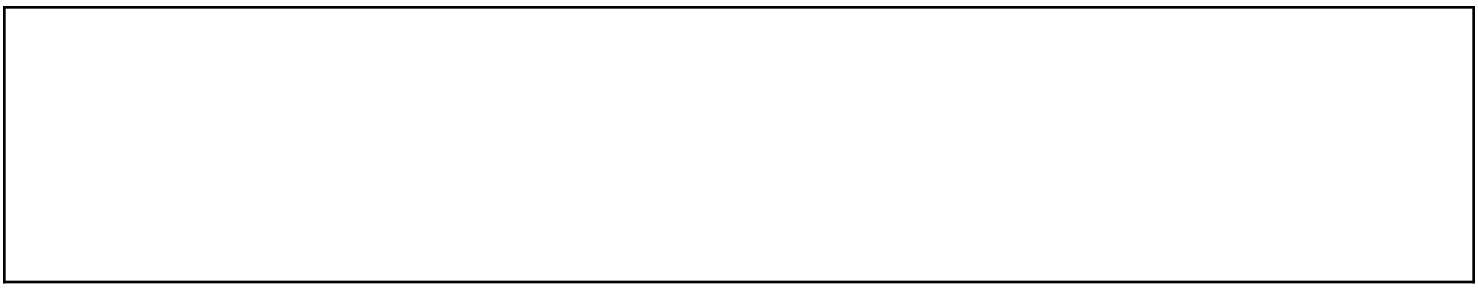
Question 3. Find $P(X \geq 3)$ with parameter 4.

Question 4. Find $P(1 \leq X \leq 4)$ with parameter 2.

Question 5. A website receives an average of 6 visits per minute. Find the probability of receiving exactly 8 visits in a minute.

Question 6. For a Poisson distribution with parameter 9, find the mean, variance, and standard deviation.

Question 7. Average calls = 3 calls/hour. Find probability of exactly 5 calls.



VITD2301: Statistical Method -I
Practical #7

Name	Harshal Ravindra Gadre	**Roll Number**	26302A3005
Class	SYIT	**Division**	A
Subject/Course	VITD2301: Statistical Method- I		
Topic	**To compute probabilities of successes in a sample drawn without replacement from a finite population using the hypergeometric distribution.**		

Question 1. A company produces a batch of 500 electronic components. Out of these, 10% are defective. If a customer orders 100 components, find the probability of receiving exactly 15 defective components.

Solution :

Q1 solution :

population size N		500
Defective components M		50
sample size n		100
Required defective components x		15
P(X=x)	HYPGOEM.DIST(x,n,M,N,False)	
P(x=15)	0.027021	

Question 2. From the same batch of 500 components with 50 defective components, a customer orders 200 components. Find the probability of receiving at least 25 defective components. Find the probability of receiving fewer than 5 defective components.

Q2. solution :	**From the same batch of 500 components with 50 defective components,**	
population size N		500
Defective components M		50
sample size n		200
P(X<=x)	HYPGOEM.DIST(x,n,M,N,True)	
P(X<=4)	HYPGOEM.DIST(4,200,50,500,True)	
	1.32E-07	
P(X>=25)	1-P(X<=24)	
	1-HYPGOEM.DIST(24,200,50,500,true)	
	0.086278	

Question 3. In a given hypergeometric distribution, Population size = 1000, Defective components = 100, Sample size = 200. Find E(X), V(X) and Standard Deviation.

population size N		1000
Defective components M		100
sample size n		200
N-M(1000-100)		900
N-n(1000-200)		800
E[X]	(n*M)/N	
	(200*100)/1000	
	20	
V[X]	(n*M)*(N-M)*(N-n)/(N^2*(N-1)	
	200*100*900*800/(1000^2*999)	
	14.41441	
SD[X]	SQRT(V[X])	
	3.796632	

Question 4. In a local election, there are 5000 registered voters. Out of these, 60% are women. A sample of 100 voters is selected. Find the probability that exactly 50 voters are women.

population size(register voters) N		5000	
sample size(voters) n		100	
out of this 60% are women		60% of 5000	3000
thus total men (M)		5000-3000	2000
for x=		50	
P(x=50)	HYPGOEM.DIST(x,n,M,N,FALSE)		
	HYPGOEM.DIST(50,100,2000,5000,FALSE)		
	0.010016		

Question 5. Using the same election data, find the probability that at least 55 sampled voters are women. Find the probability that less than 40 sampled voters are women.

population size(register voters) N		5000	
sample size(voters) n		100	
out of this 60% are women		60% of 5000	3000
thus total men (M)		5000-3000	2000
for x=		50	
P(X>=55)	1-P(X<=54)		
	1-HYPGEOM.DIST(54,100,3000,5000,TRUE)		
	0.871298		
P(X>=39)	1-P(X<=39)		
	HYPGEOM.DIST(39,100,3000,5000,TRUE)		
	1.49E-05		

VITD2301 : Statistical Method -I
Practical 7 – Practice Exercise

Name		**Roll Number**	
Class	SYIT	**Division**	
Subject/Course	VITD2301 : Statistical Method- I		

Topic	To compute probabilities of successes in a sample drawn without replacement from a finite population using the hypergeometric distribution.

Question 1. A box contains 40 bulbs, out of which 8 are defective. If 5 bulbs are selected randomly, find the probability that exactly 2 are defective.

Question 2. A box contains 40 bulbs, out of which 8 are defective. If 5 bulbs are selected randomly, find the probability that at most 1 bulb is defective.

Question 3. In a class of 60 students, 20 students have enrolled for Python. If 10 students are selected randomly, find the probability that exactly 4 selected students have enrolled for Python.

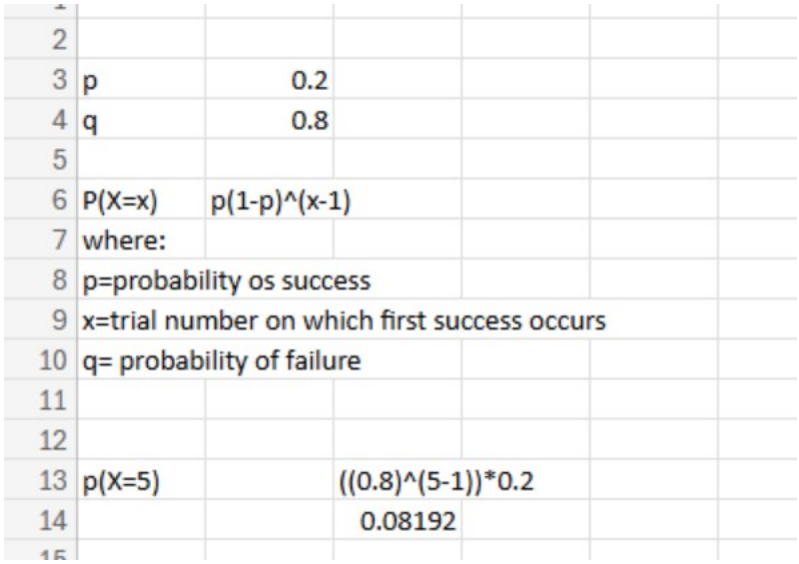
Question 4. In a class of 60 students, 20 students have enrolled for Python. If 10 students are selected randomly, find the probability that at least 3 selected students have enrolled for Python.

VSIT | Vidyalankar School of Information Technology
RE-ACCREDITED : GRADE 'A' BY NAAC

VITD2301: Statistical Method -I
Practical #8

Name	*Harshal Ravindra Gadre*	**Roll Number**	*26302A3005*
Class	SYIT	**Division**	*A*
Subject/Course	VITD2301: Statistical Method- I		
Topic	**To determine the probability of the first success occurring on a specific trial using the geometric distribution, enabling analysis of event occurrences over repeated independent trials in Excel.**		

Question 1. A call centre receives many customer service calls. On average, 20% of calls are escalated to a supervisor. Find the probability that the first escalated call occurs on the 5th call.



2			
3	p	0.2	
4	q	0.8	
5			
6	P(X=x)	p(1-p)^(x-1)	
7	where:		
8	p=probability os success		
9	x=trial number on which first success occurs		
10	q= probability of failure		
11			
12			
13	p(X=5)		((0.8)^(5-1))*0.2
14			0.08192

Question 2. Using the same call centre data, find the probability that the first escalated call occurs within

the first 3 calls. Find the probability that the first escalated call occurs after the 7th call.

1		
2		
3	p	0.2
4	q	0.8
5		
6		
7	P(X=x)	1-(1-p)^x
8	P(X<=3)	1-0.8^3
9		0.488
10		
11	P(X>7)	1-P(X<=7)
12		1-(1-0.8^7)
13		0.8^7
14		0.209715
15		
16		

Question 3. In a given geometric distribution, Probability of Success p=0.3 . Find E(X), V(X) and Standard Deviation.

2		
3	p	0.3
4	q	0.7
5		
6	E(X)	1/p
7		1/0.3
8		3.333333
9		
10	V(X)	1-p/p^2
11		0.7/(0.3^2)
12		7.777777
13		
14	SD(X)	sqrt(V(X))
15		2.788867
16		

Question 4. A tech support team receives customer emails. Based on past data, 15% of emails contain urgent issues. Find the probability that the first urgent email is the 8th one checked.

1			
2			
3			
4	p	0.15	
5	1-p	0..85	
6			
7			
8	P(X=x)		(q^(x-1))*p
9	P(X=8)		(0.85^(8-1))*0.15
10			0.048087
11			
12			
13			
14			

Question 5. Using the same tech support data, find the probability that the first urgent email is received within the first 5 emails. Find the probability that the first urgent email is received after checking 10

emails.

1		
2	p	0.15
3	q	0.85
4		
5		
6	P(X=x)	1-q^x
7	P(X=5)	1-0.85^5
8		0.556295
9		
10		
11	P(X>10)	1-P(X<=10)
12		1-(1-0.85^10)
13		0.85^10
14		0.196874
15		
16		

VSIT | Vidyalankar School of Information Technology
RE-ACCREDITED : GRADE 'A' BY NAAC

VITD2301 : Statistical Method -I
Practical 8 – Practice Exercise

Name		**Roll Number**	
Class	SYIT	**Division**	
Subject/Course	VITD2301 : Statistical Method- I		
Topic	**To determine the probability of the first success occurring on a specific trial using the geometric distribution, enabling analysis of event occurrences over repeated independent trials in Excel.**		

Question 1. A salesperson has a 30% chance of making a successful sale on each call. Find the probability that the first sale happens on the 4th call.

Question 2. A salesperson has a 30% chance of making a successful sale on each call. Find the probability that the first sale happens within the first 3 calls.

Question 3. A machine has a 10% chance of producing a defective item. Find the probability that the first defective item is produced on the 6th item.

Question 4. A machine has a 10% chance of producing a defective item. Find the probability that the first defective item occurs after 5 items.

VITD2301 : Statistical Method -I
Practical #11

Name	Harshal Ravindra Gadre	**Roll Number**	26302A3005
Class	SYIT	**Division**	A
Subject/Course	VITD2301 : Statistical Method- I		
Topic	**To analyze relationships between variables and predict sales using correlation and regress Python.**		

Question 1. Generate a random dataset for sales analysis with 100 observations including Advertising Spend, Pr the basic information and summary statistics of the dataset.

Code:

```python
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

np.random.seed(0)

n = 100
advertising = np.random.randint(1000, 5000, n)
price = np.random.randint(50, 300, n)

sales = 50 + 0.5 * advertising - 0.3 * price + np.random.normal(0, 10, n)

# Pass the dictionary directly into pd.DataFrame(...)
data = pd.DataFrame({
    "Advertising": advertising,
    "Price": price,
    "Sales": sales,
})

print(data.head())
print(data.info())
print(data.describe())
print(data.isnull().sum())
print(data.corr())
print(data.tail())
```

Output:

```
   Advertising  Price        Sales
0         3732    118  1885.256624
1         3607     56  1821.337563
2         2653    246  1317.582522
3         4264     97  2171.858892
4         1835    177   926.187796
<class 'pandas.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 3 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   Advertising  100 non-null    int32
 1   Price        100 non-null    int32
 2   Sales        100 non-null    float64
dtypes: float64(1), int32(2)
memory usage: 1.7 KB
None
       Advertising       Price        Sales
count   100.000000  100.000000   100.000000
mean   3085.050000  162.870000  1545.440881
std    1083.839699   69.435917   541.638656
min    1084.000000   50.000000   517.589744
25%    2018.750000   98.000000  1019.535481
50%    3199.000000  164.500000  1610.967321
75%    3906.000000  225.000000  1951.567610
max    4942.000000  294.000000  2512.180298
Advertising    0
Price          0
Sales          0
dtype: int64
             Advertising     Price     Sales
Advertising     1.000000  0.043620  0.999144
Price           0.043620  1.000000  0.006609
Sales           0.999144  0.006609  1.000000
    Advertising  Price        Sales
95         2589     74  1315.404502
96         3956    175  1970.944675
97         3681     52  1875.074792
98         1084     53   572.560061
99         3251    144  1618.550487
```

Question 2. Check for any missing values in the dataset. Compute the correlation matrix between all variables.

Code:

```python
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

np.random.seed(0)

n = 100
advertising = np.random.randint(1000, 5000, n)
price = np.random.randint(50, 300, n)

sales = 50 + 0.5 * advertising - 0.3 * price + np.random.normal(0, 10, n)

# Pass the dictionary directly into pd.DataFrame(...)
data = pd.DataFrame({
    "Advertising": advertising,
    "Price": price,
    "Sales": sales,
})

print(data.head())
print(data.info())
print(data.describe())
print(data.isnull().sum())
print(data.corr())
print(data.tail())

sns.heatmap(data.corr(), annot=True)
plt.title("Correlation Heatmap")
plt.show()

plt.scatter(data['Advertising'], data['Sales'])
plt.xlabel("Advertising")
plt.ylabel("Sales")
plt.title("Advertising vs Sales")
plt.show()
```

Output:

```
   Advertising  Price        Sales
0         3732    118  1885.256624
1         3607     56  1821.337563
2         2653    246  1317.582522
3         4264     97  2171.858892
4         1835    177   926.187796
<class 'pandas.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 3 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   Advertising  100 non-null    int32
 1   Price        100 non-null    int32
 2   Sales        100 non-null    float64
dtypes: float64(1), int32(2)
memory usage: 1.7 KB
None
       Advertising       Price        Sales
count   100.000000  100.000000   100.000000
mean   3085.050000  162.870000  1545.440881
std    1083.839699   69.435917   541.638656
min    1084.000000   50.000000   517.589744
25%    2018.750000   98.000000  1019.535481
50%    3199.000000  164.500000  1610.967321
75%    3906.000000  225.000000  1951.567610
max    4942.000000  294.000000  2512.180298
Advertising    0
Price          0
Sales          0
dtype: int64
              Advertising     Price     Sales
Advertising      1.000000  0.043620  0.999144
Price            0.043620  1.000000  0.006609
Sales            0.999144  0.006609  1.000000
    Advertising  Price        Sales
95         2589     74  1315.404502
96         3956    175  1970.944675
97         3681     52  1875.074792
98         1084     53   572.560061
99         3251    144  1618.550487
```

Question 3. Visualize the correlation matrix using a heatmap. Plot a scatter graph between Advertising Spend and Sales.

Code:

```python
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

np.random.seed(0)

n = 100
advertising = np.random.randint(1000, 5000, n)
price = np.random.randint(50, 300, n)

sales = 50 + 0.5 * advertising - 0.3 * price + np.random.normal(0, 10, n)

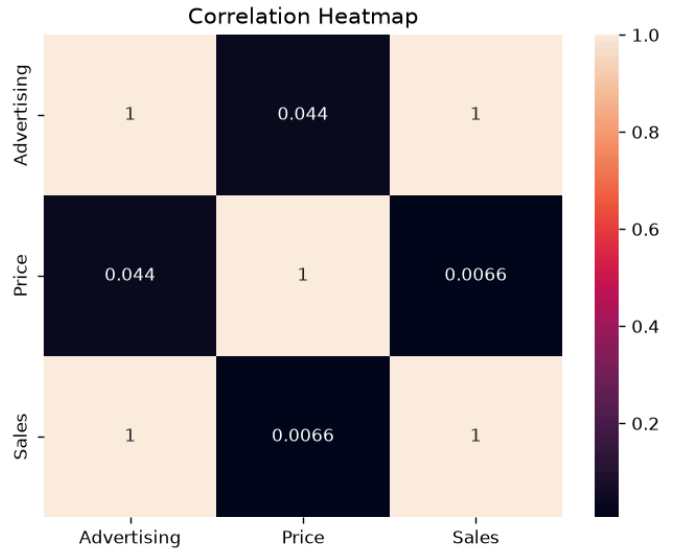
# Pass the dictionary directly into pd.DataFrame(...)
data = pd.DataFrame({
    "Advertising": advertising,
    "Price": price,
    "Sales": sales,
})

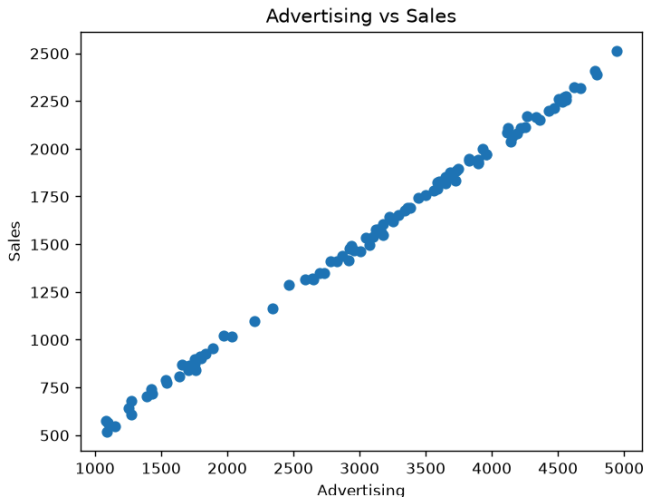
print(data.head())
print(data.info())
print(data.describe())
print(data.isnull().sum())
print(data.corr())
print(data.tail())

sns.heatmap(data.corr(), annot=True)
plt.title("Correlation Heatmap")
plt.show()

plt.scatter(data['Advertising'], data['Sales'])
plt.xlabel("Advertising")
plt.ylabel("Sales")
plt.title("Advertising vs Sales")
plt.show()
```

Output:





Question 4. Plot a scatter graph between Price and Sales. Fit a simple linear regression model using Advertising Spend to predict Sales.

Code:

```python
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

np.random.seed(0)

n = 100
advertising = np.random.randint(1000, 5000, n)
price = np.random.randint(50, 300, n)

sales = 50 + 0.5 * advertising - 0.3 * price + np.random.normal(0, 10, n)

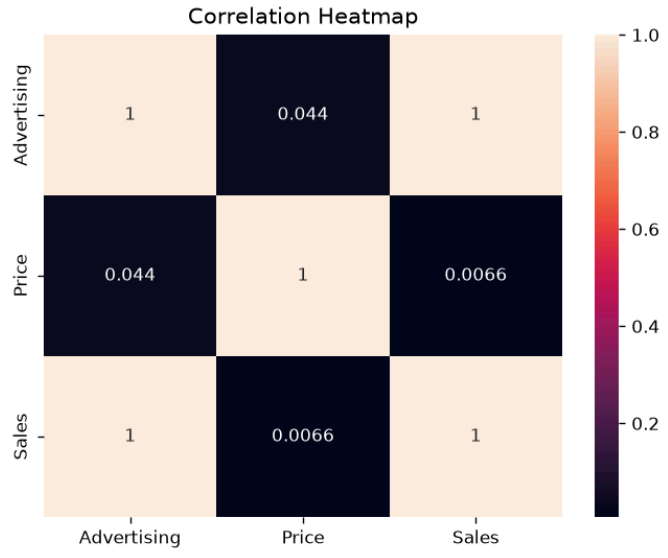
# Pass the dictionary directly into pd.DataFrame(...)
data = pd.DataFrame({
    "Advertising": advertising,
    "Price": price,
    "Sales": sales,
})

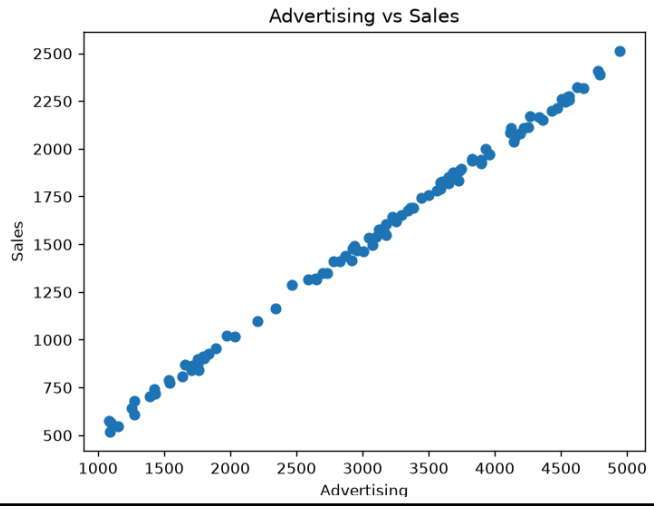
print(data.head())
print(data.info())
print(data.describe())
print(data.isnull().sum())
print(data.corr())
print(data.tail())

sns.heatmap(data.corr(), annot=True)
plt.title("Correlation Heatmap")
plt.show()

plt.scatter(data['Advertising'], data['Sales'])
plt.xlabel("Advertising")
plt.ylabel("Sales")
plt.title("Advertising vs Sales")
plt.show()
```

Output:





Question 5. **Plot the regression line for Advertising Spend vs Sales. Fit a multiple linear regression model using Advertising Spend and Price to predict Sales and display the coefficients.**

```python
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

np.random.seed(0)

n = 100
advertising = np.random.randint(1000, 5000, n)
price = np.random.randint(50, 300, n)

sales = 50 + 0.5 * advertising - 0.3 * price + np.random.normal(0, 10, n)

# Pass the dictionary directly into pd.DataFrame(...)
data = pd.DataFrame({
    "Advertising": advertising,
    "Price": price,
    "Sales": sales,
})

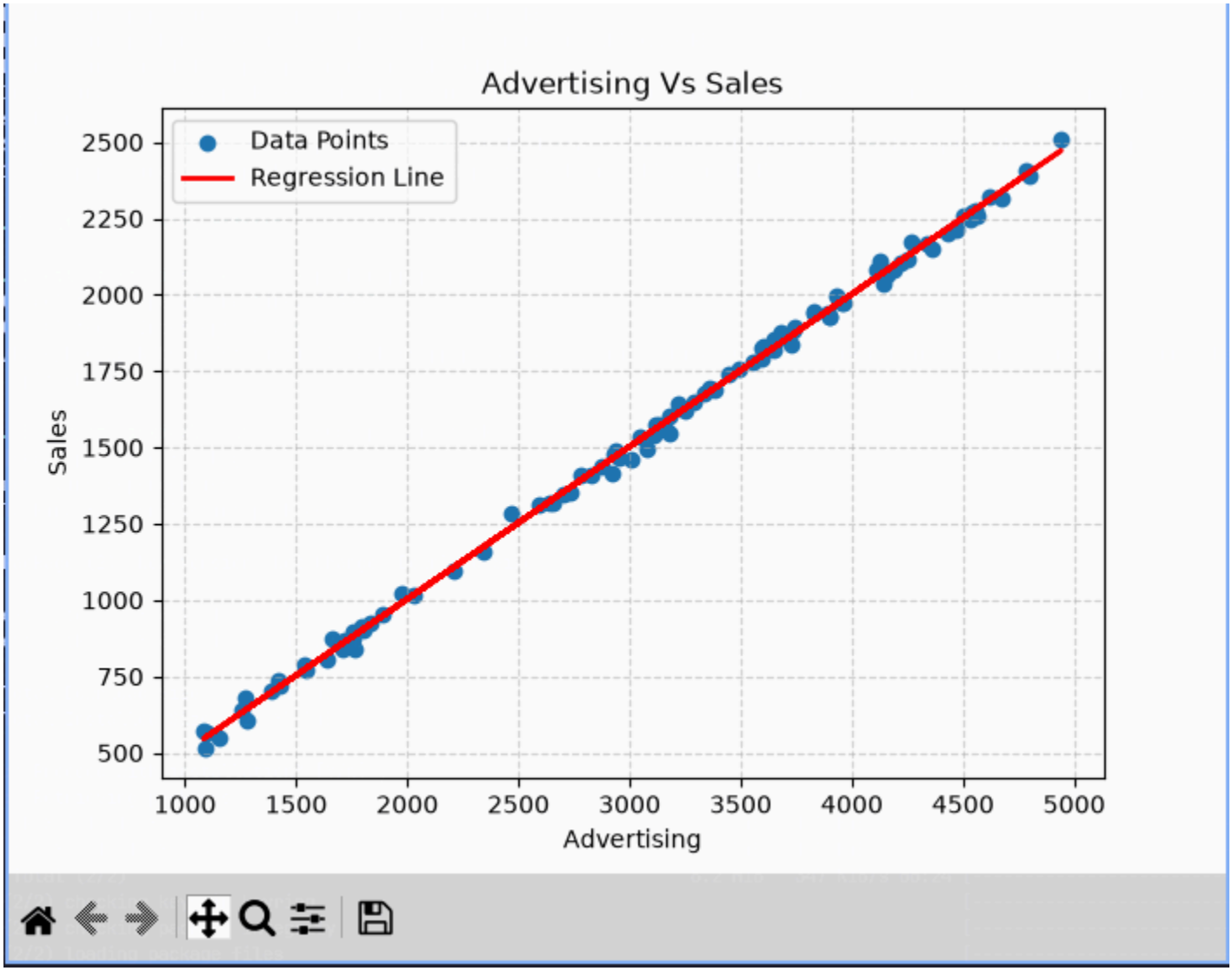
print(data.head())
print(data.info())
print(data.describe())
print(data.isnull().sum())
print(data.corr())
print(data.tail())

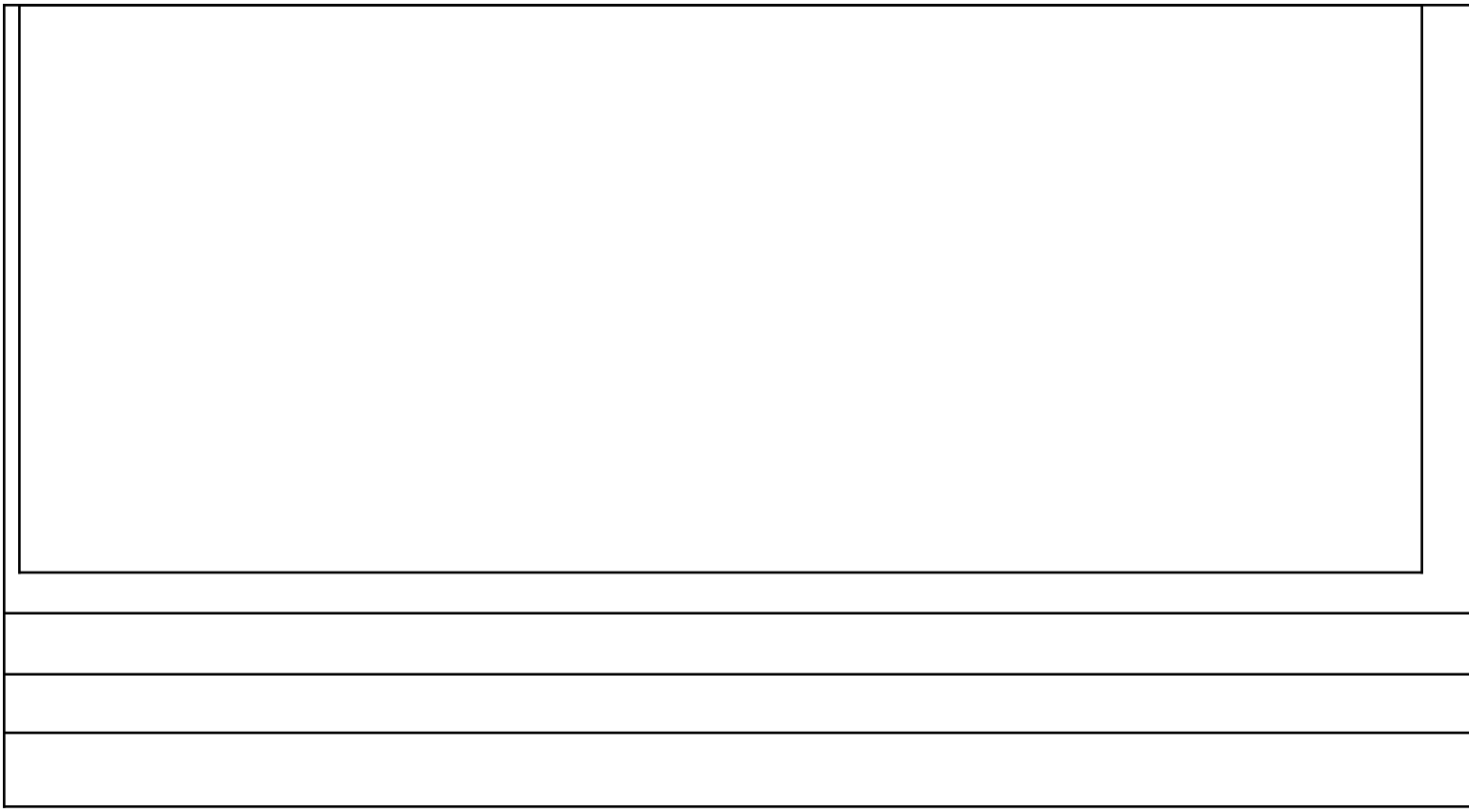
sns.heatmap(data.corr(), annot=True)
plt.title("Correlation Heatmap")
plt.show()

plt.scatter(data['Advertising'], data['Sales'])
plt.xlabel("Advertising")
plt.ylabel("Sales")
plt.title("Advertising vs Sales")
plt.show()

X = data[['Advertising', 'Price']]
y = data['Sales']
# Fit the multiple linear regression without requiring scikit-learn.
X_with_intercept = np.column_stack((np.ones(len(X)), X.to_numpy()))
coefficients = np.linalg.lstsq(X_with_intercept, y.to_numpy(), rcond=None)[0]
print(f"Intercept: {coefficients[0]}")
print(f"Coefficients: {coefficients[1:]}")
```

Output:


Advertising Vs Sales
Data Points
Regression Line
Sales
Advertising
2500
2250
2000
1750
1500
1250
1000
750
500
1000
1500
2000
2500
3000
3500
4000
4500
5000



S={(1,1),(1,2),(1,3),(1,4),(1,5),(1,6),(2,1),(2,2),(2,3),(2,4),(2,5),(2,6),(3,1),(3,2),(3,3),(3,4),(3,4),(3,5),(3,6),(

n(S) 36

A={(1,6),(2,5),(3,4),(4,3),(5,2),(6,1),(5,6),(6,5)}

n(A) 8

P(A) 0.222222

B={(4,4),(4,5),(4,6),(5,4),(5,5),(5,6),(6,4),(6,5),(6,6)}

n(B) 9

P(B) 0.25

C={(1,4),(2,4),(3,4),(4,4),(5,4),(6,4)}

n© 6

P© 0.166667

A={(2,6),6,2),(3,5),(5,3),(4,4)}

n(A) 5

B={(5,1),(5,2),(5,3),(5,4),(5,5),(5,6)}

n(B) 6

A∩B={(5,3)} n(A∩B) 1

P(B) 0.166667

P(A∩B) 0.027778

P(A|B) 0.166667

4,1),(4,2),(4,3),(4,4),(4,5),(4,6),(5,1),(5,2),(5,3),(5,4),(5,5),(5,6),(6,1),(6,2),(6,3),(6,4),(6,5),(6,6)

n(S)	36
A = { (4,4),(4,5),(4,6),(5,5),(5,6),(6,6)}	
n(A)	9
P(A) = n(A)/n(S)	0.25

n(S)	36
B: numbers less than equal to 4	
B = {(1,1),..,(1,4),(2,1),..,(2,4),(3,1),..,(3,4),(4,1),..,(4,4)}	
n(B)	16
P(B) = n(B)/n(S)	0.444444

n(S)	36
C ={(1,6),(2,6),(3,6),(4,6),(5,6),(6,1),(6,2),(6,3),(6,4),(6,5),(6,6)}	
n(C)	11
P(C) = n(C) / n(S)	
P(C)	0.305556

n(S)	36
D={(2,1),(2,3),(2,5),(4,1),(4,3),(4,5),(6,1),(6,3),(6,5)}	
n(D)	9
P(D) = n(D) / n(S)	
P(D)	0.25

n(A)	400
n(B)	350
n(A ∩ B)	200
P(B | A)	0.5
P(A | B)	0.571429
P(A ∩ B)	550

P(D)	0.05
P(R)	0.02
P(D ∩ R)	0.01

P(D ∩ R)

P(D) + P(R) - P(D ∩ R)	0.06

P(ND ∩ NR)

P(ND) + P(NR) - P(ND ∩ NR)

X/Y	Y	Y	Y
X/Y	0	1	2
1	0.15	0.3	0.15
2	0.15	0.05	0.2

Total Probability	1
P(X=1)	0.6
P(X=2)	0.4
P(Y=0)	0.3
P(Y=1)	0.35
P(Y=2)	0.35

X/Y	Y	Y	Y
X/Y	0	1	2
1	0.15	0.3	0.15
2	0.15	0.05	0.2

Total Probability	1
P(X=1)	0.6
P(X=2)	0.4
P(Y=0)	0.3
P(Y=1)	0.35
P(Y=2)	0.35

E[X] = SUM * Probability of X

E[X] 1.4

E[Y] = SUM * Probability of Y

E[Y] 1.05

X/Y	Y	Y	Y
X/Y	0	1	2
1	0.15	0.3	0.15
2	0.15	0.05	0.2

Total Probability	1
P(X=1)	0.6
P(X=2)	0.4
P(Y=0)	0.3
P(Y=1)	0.35
P(Y=2)	0.35

E[XY]	1.5

X/Y	Y	Y	Y
X/Y	0	1	2
1	0.15	0.3	0.15
2	0.15	0.05	0.2

Total Probability	1
P(X=1)	0.6
P(X=2)	0.4
P(Y=0)	0.3
P(Y=1)	0.35
P(Y=2)	0.35

E[X] = SUM * Probability of X

E[X]	1.4

E[Y] = SUM * Probability of Y

E[Y]	1.05

E[X^2]	2.2
X[X]^2	1.96
Var[X]	0.24

E[Y^2]	1.75
E[Y]^2	1.1025
Var[X]	0.6475

X/Y	Y	Y	Y
X/Y	0	1	2
1	0.15	0.3	0.15
2	0.15	0.05	0.2

Total Probability	1
P(X=1)	0.6
P(X=2)	0.4
P(Y=0)	0.3
P(Y=1)	0.35
P(Y=2)	0.35
E[X] = SUM * Probability of X	
E[X]	1.4
E[Y] = SUM * Probability of Y	
E[Y]	1.05
E[X^2]	2.2
X[X]^2	1.96
Var[X]	0.24
E[Y^2]	1.75
E[Y]^2	1.1025
Var[Y]	0.6475
E[XY]	1.5
Cov (X,Y)	0.03
Correlation Coefficient	0.076102

n =	5
p =	0.4
1 - p	0.6
x <= 2 =	3
x > 2 =	1

PMF	0.2304
CDF P(X <= 2)	0.91296
CDF P(X > 2)	0.33696
CDF P(X >= 2	0.66304

n =		10
p =		0.5
1 - p =		0.5

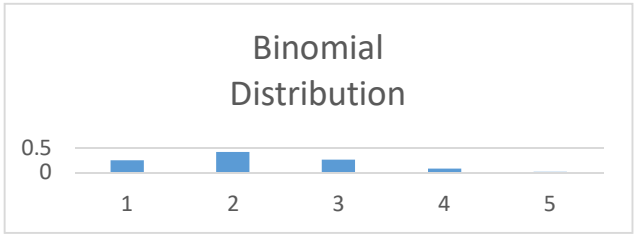
Expectation of X =

X ~ BINOMIAL =	N*p	
E[X]		5

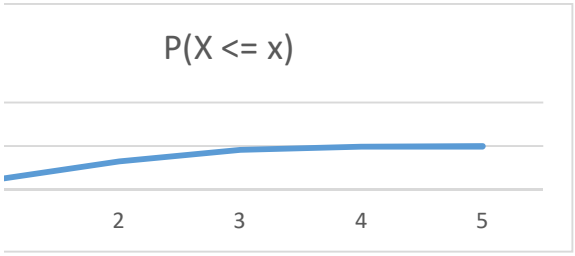
n =	20
p =	0.35
1 - p =	0.65
Var (X)	4.55
SD (X)	2.133073

n =	4
p =	0.3
1 - p =	0.7

X = x	P(X = x)	P(X <= x)
0	0.2401	0.2401
1	0.4116	0.6517
2	0.2646	0.9163
3	0.0756	0.9919
4	0.0081	1

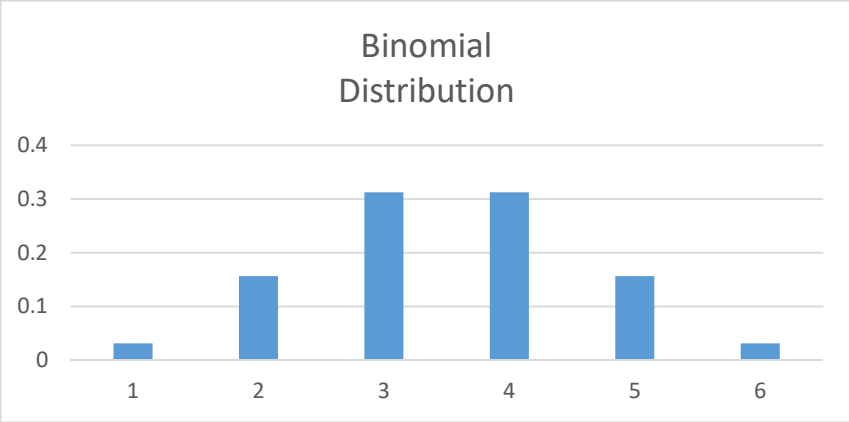


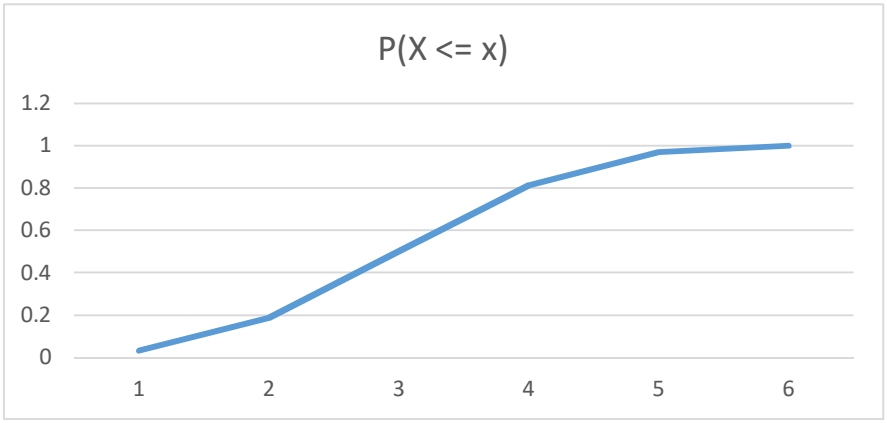



P(X <= x)
2
3
4
5

n =	5
p =	0.5
1 - p =	0.5

X = x	P(X = x)	P(X <= x)
0	0.03125	0.03125
1	0.15625	0.1875
2	0.3125	0.5
3	0.3125	0.8125
4	0.15625	0.96875
5	0.03125	1




P(X <= x)
1.2
1
0.8
0.6
0.4
0.2
0
1
2
3
4
5
6